

Wen-Kai "SansWord" Huang

Senior Software Engineer · Data Infrastructure · Distributed Systems · AI/ML Platform

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"A data platform builder interested in AI systems and their mathematical foundations."



sansword.github.io/resume

Scan for full resume & projects

PROFESSIONAL SUMMARY

Senior Software Engineer with **9+ years** designing and operating large-scale distributed systems and global data infrastructure on **AWS and GCP**, serving **~900M monthly active users** across US, APAC, and Europe. Specialized in high-throughput, fault-tolerant pipelines (>20K msgs/sec), scalable data platforms, and cloud-native migrations. Proven technical leader recognized for cross-functional collaboration and large-scale impact. Currently exploring **agentic development** and harness engineering through hands-on projects and public writing.

PROFESSIONAL EXPERIENCE

Yahoo!

Sr. Software Dev Engineer · Sunnyvale, CA · Feb 2019 – Jun 2025

- Led end-to-end delivery of Yahoo's data infrastructure evolution to GCP, leading a 5+ engineer team to build a GCP-native ingestion platform using **Cloud Composer, Airflow, and Dataproc** — improving data freshness from 3+ hours to **under 60 minutes** across US, APAC, and Europe.
- Supported a **zero-downtime AWS migration** with Kubernetes-based auto-scaling, maintaining multi-region lambda architecture serving recommendation, ads, and downstream systems across News, Sports, and Finance.
- Built aggregation and caching layers that **reduced data scanned from 725 GB to 121 KB**, cutting query costs by **70%** on Yahoo's BigQuery infrastructure.
- Drove on-call reliability initiative, reducing open incident tickets **from 1,000+ to zero**; practices adopted by the broader on-call team.
- Co-invented a patented salient entity algorithm ([US11803605B2](#)) integrating ranking models into production recommendation systems.

Yahoo! E-Commerce Platform

Sr. Engineer · Taipei, Taiwan · Jun 2016 – Feb 2019

- Drove early AWS adoption across Yahoo engineering, building **blue/green deployment** and auto-scaling patterns via ECS, later adopted by global teams.
- Contributed to migrating **80 TB** to S3 with zero downtime using dual-write/dual-read patterns to ensure safe cutover.
- Scaled **Pigeon** (Apache Pulsar message bus) to **20K msgs/sec**; improved MTBF from 3.5 days to **90 days (~26x)**.
- Core developer of **Cupid**, a multi-tenant discount service with a JSON-based rule engine — driving **10K/day coupon redemption growth**.

PATENT

Determining Salient Entities and Generating Salient Entity Tags Based Upon Articles

[US11803605B2](#)

Co-invented methods to identify salient entities at scale for production recommendation systems at Yahoo.

EDUCATION

M.S., Computer Science

Programming Languages and Software Methodology
National Chengchi University, Taipei, Taiwan

B.S., Mathematics

National Chengchi University, Taipei, Taiwan

TECHNICAL SKILLS

Languages	Python, Java, SQL, Shell, Groovy
Cloud – GCP	Dataproc, Cloud Composer, Dataflow, Airflow, BigQuery
Cloud – AWS	S3, ECS, ELB, ElastiCache
Infrastructure	Terraform, Kubernetes, Docker, Redis
Stream Processing	Apache Storm, Apache Pulsar, Kafka
AI Practice	Agentic Programming, Harness Engineering
Writing	Technical blog posts, study guides

SIDE PROJECTS & WRITING

My Tiny AI Bootcamp

3 weeks AI skeptic → shipping 3 projects with Claude. Reflects on where senior judgment steers AI-assisted design decisions.

Leaderboard System Design — Study Guide

Full system design (Redis sorted sets, fan-out, sharding) with companion note on AI for interview prep.

sans_learning

Measured notes: 7.8x cost delta between Sonnet 4.6 and Opus 4.7; per-story-point cost analysis; harness engineering.

sans_cube

Real-time Rubik's cube analyzer — BLE ingestion, live 3D rendering, CFOP/Roux phase detection, Firestore sync.